

Full Stack Development - Dr. Angela Yu



Sec 1 - FED

nslookup.io - shows exact ip address of searched servers.

submarinecablemap.com - shows undersea cables full of optic fibers.



People generally do - Code along

Recommendation - Try to understand the 10 minutes, now replicate the 10 minutes. Forces you to actually understand what is going on.

keybr.com - Learn to type faster.

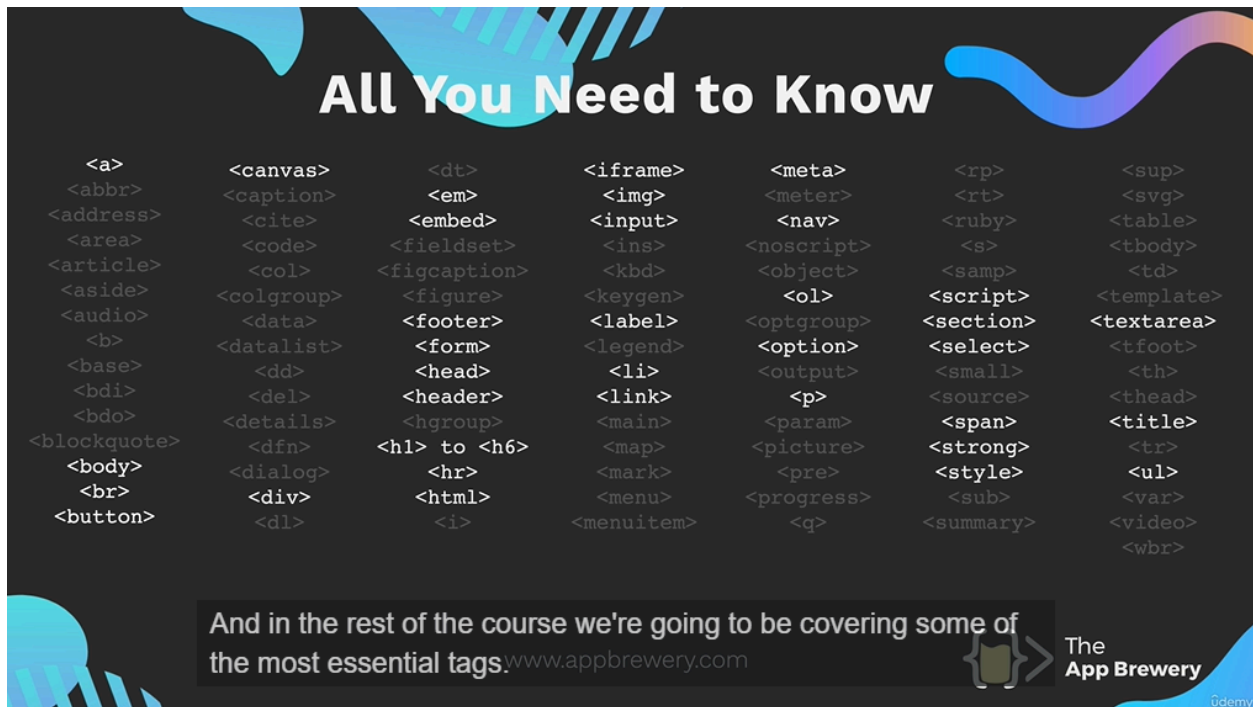
Practice = Mastery

When Stuck - Try stack overflow website.

Screenshot is Windows: **Windows + Shift + S** (part of screen)

Sec 2 - Intro to Html

HTML - Hypertext Markup Language



<h1>Heading 1</h1> -> Biggest

<h2>Heading 2</h2>

<h3>Heading 3</h3>

<h4>Heading 4</h4>

<h5>Heading 5</h5>

<h6>Heading 6</h6> -> Smallest

[If you don't understand something, google it.]

<p>Paragraph</p> - Paragraph tag

[lipsum.com](https://www.lipsum.com/) - Lorem ipsum dolor sit amet

<hr /> - Horizontal rule tag -> Void element. Does Not need a closing tag.

 - Break element

[diffchecker.com](https://www.diffchecker.com/) - Checks the difference between your code and the answer.

Sec 3 - Inter Html

Lists _

Unordered List -

Milk

Eggs

Flour

→ It will give you bullet points.

Ordered List -

```
<ol>
  <li>Milk</li>
  <li>Eggs</li>
  <li>Flour</li>
</ol>
```

→ It will give you numbers.

<a> → Anchor Element.

Ex: `<a href="http://www.google.com">This is a link</a>`

HTML Attributes:

`<tag attribute=value>Content</tag>`

`<tag attribute=value another attribute=value>Content</tag>` → Should be in the opening tag.

Ex: `<a draggable=true> This is a link to Google </a>` → We can drag the text.

The image tag:

`<img src = "url"/>` → A self closing tag.

picsum.photos - Lorem Picsum

`<img src = "https://picsum.photos/200" alt="forest at sunset"/>`

(Also revise from offline)

Sec 4 - Multi-page Websites

File path:

1. Absolute File path - Relative to the root of the computer.

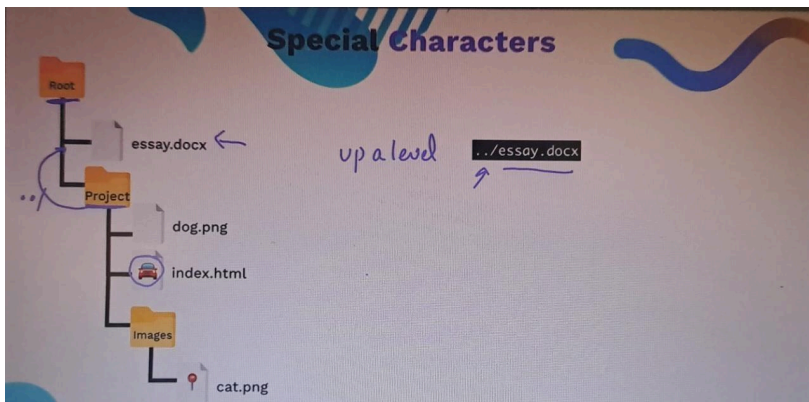
Ex: C:/Project/Image/cat.png

2. Relative File Path - Relative to a file.

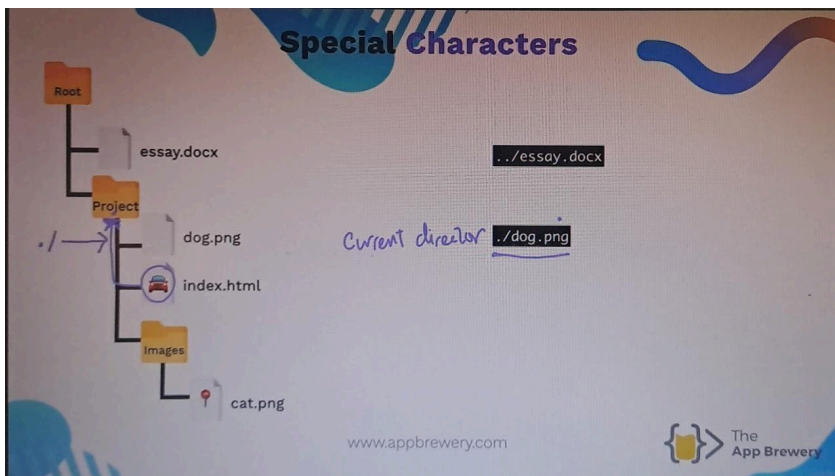
Suppose we take Project folder as ref, where index.html lies, then_

Path: Image/cat.png

Special Characters -

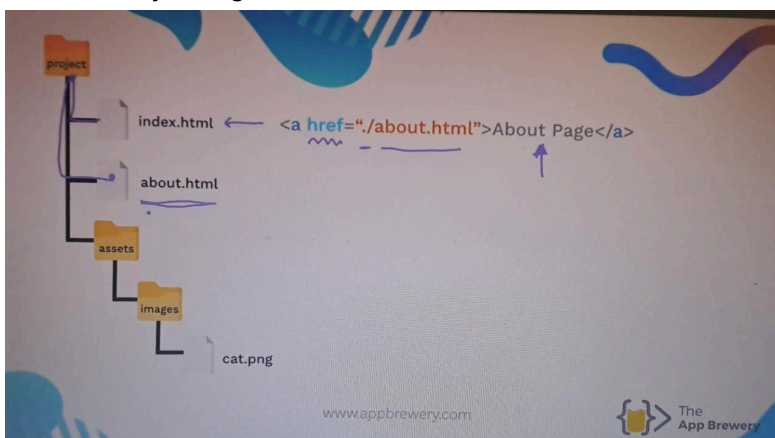


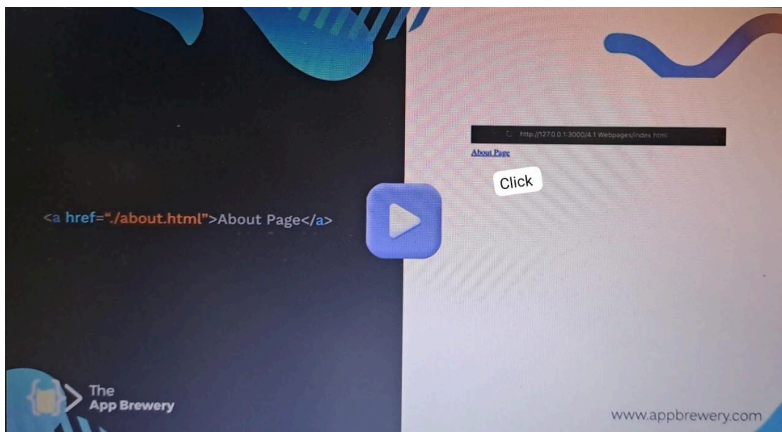
→ ../essay.docx - up a level



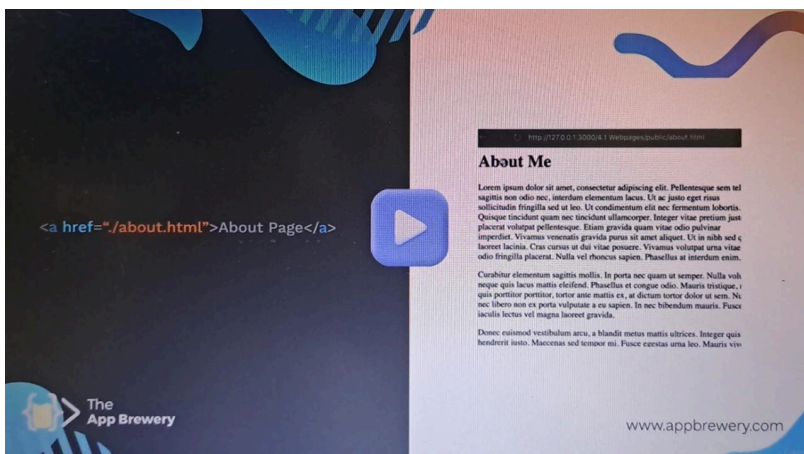
→ ./dog.png - current level

Now, let's try doing that with html files.





→ When clicked, it directs to the about.html



→ about.html is opened.

HTML - Boilerplate



Use '!' and click enter to get boilerplate in VS code.

<https://appbrewery.github.io/capstone-1-example/> - have a glance, try to make something like this.

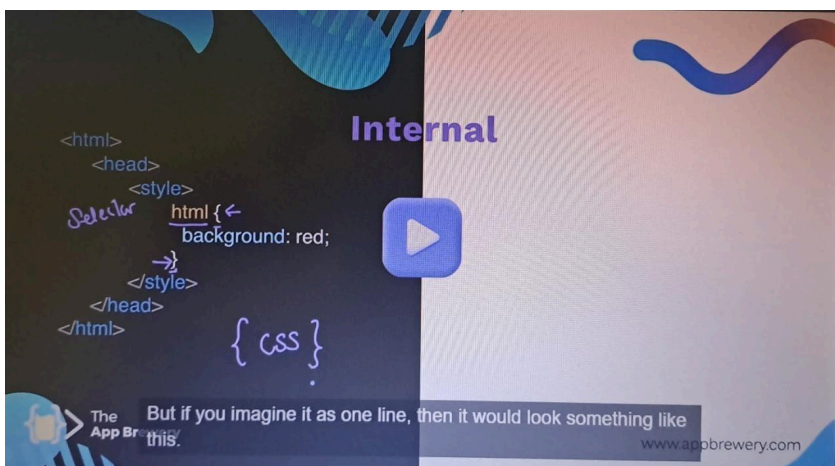
Sec 4 - Intro to CSS (Cascading Style Sheets)

<center> → Html tags for styling

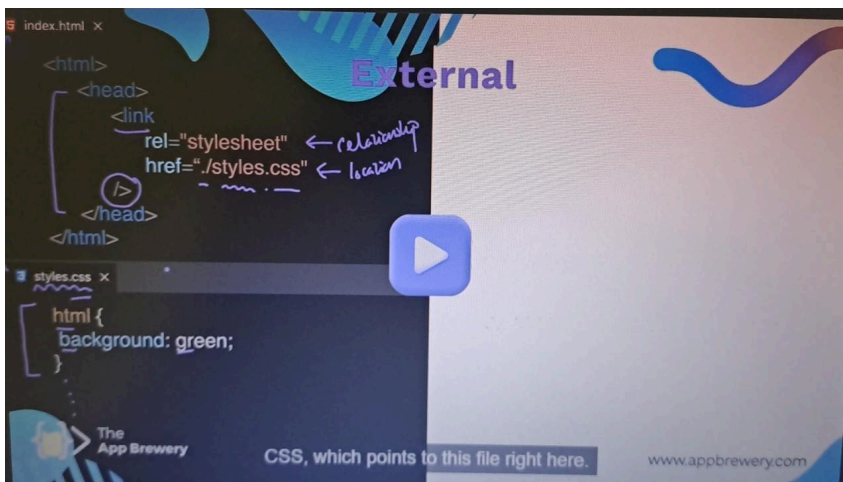
Types of adding CSS -

1. Inline: <html style= "background: blue;" > </html>

2. Internal:



3. External:



Css - property **color** → used to change text color.

Class Selector:

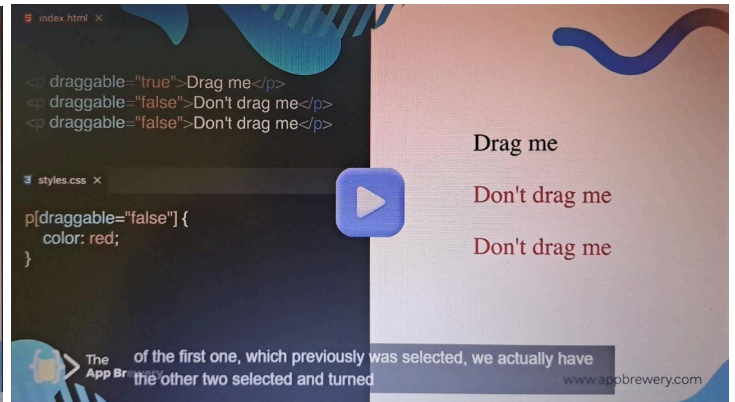
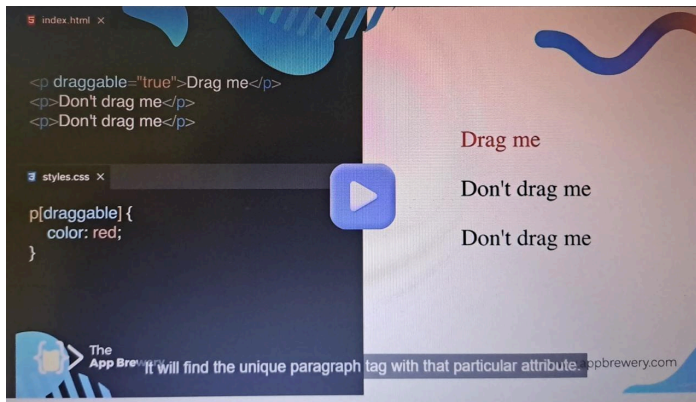
In html → <h1 class="class_head">Hello</h1>

In css → .class_head{
 Color : red;
}

Id Selector:

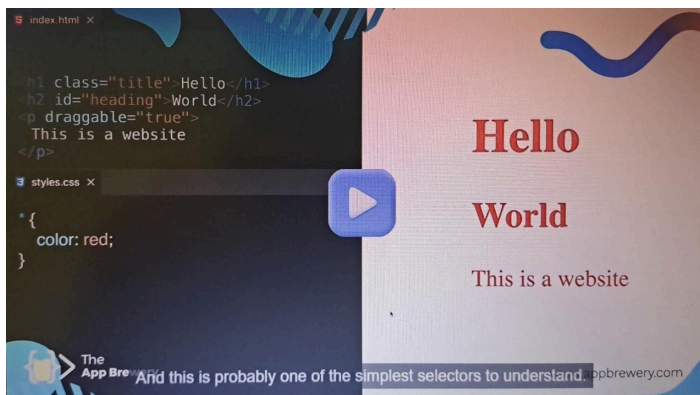
In html → <h1 id="my_heading">The Diamond</h1>
In css → #my_heading{
 Color: blue;
}

Attribute Selector:



For more clarity check 34 vid assignments.

Universal Selector:



→ To change dimensions of an image_ In css_ height: 200px; width: 200px;

Sec 4 - CSS Properties

Css colors:

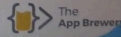
<https://colorhunt.co/> - for finding colors

<https://www.csfieldguide.org.nz/en/interactives/rgb-mixer/> - RGB mixer

Color Properties

```
html {  
  background-color: red;  
}  
  
h1 {  
  color: blue;  
}
```

Well, how do we know that we can use these colors?



Font Properties:

Font Size

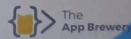
Pixel
1px

1/96th inch
0.26mm

Font 12 pt
Point
1pt

1/72nd inch
0.35mm

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Font Size

1px

1/96th inch

1pt

1/72nd inch

1em

2x 20px = 40px

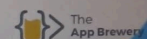
100% of parent

Relative

1rem

100% of root

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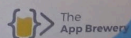
1em

100% of parent

```
<!DOCTYPE html>  
<html lang="en">  
  
<head>  
  <meta charset="UTF-8">  
  <title>Document</title>  
  <link rel="stylesheet" href="/style.css" />  
</head>  
  
<body>  
  <h1>Hello</h1>  
  <h2>Hello</h2>  
</body>  
</html>
```

2em x 20px = 40px
x 80 = 160px

www.appbrewery.com



1rem

100% of root

1em

100% of parent

???

```
<!DOCTYPE html>  
<html lang="en">  
  
<head>  
  <meta charset="UTF-8">  
  <title>Document</title>  
  <link rel="stylesheet" href="/style.css" />  
</head>  
  
<body>  
  <h1>Hello</h1>  
  <h2>Hello</h2>  
</body>  
</html>
```

90px
20px → 80px
2rem

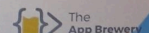
Root

www.appbrewery.com

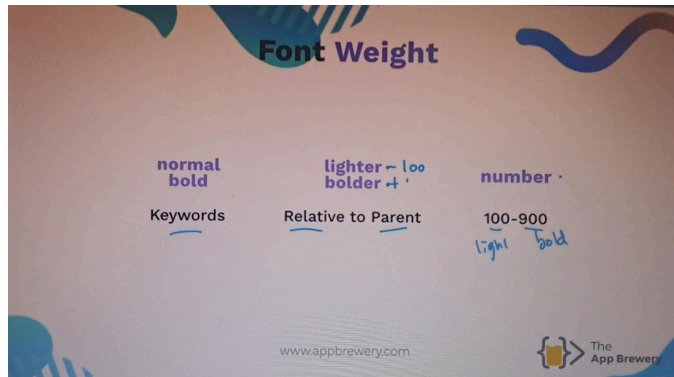
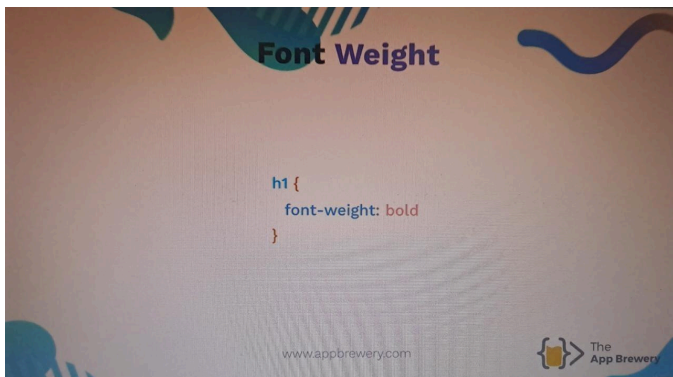
1rem

100% of root

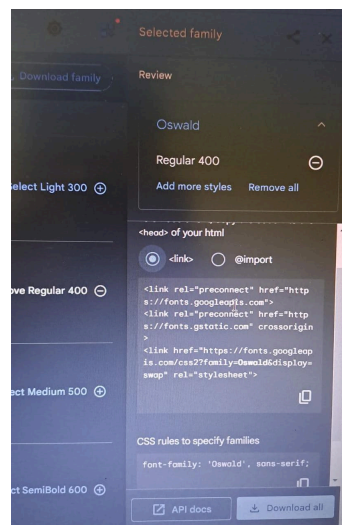
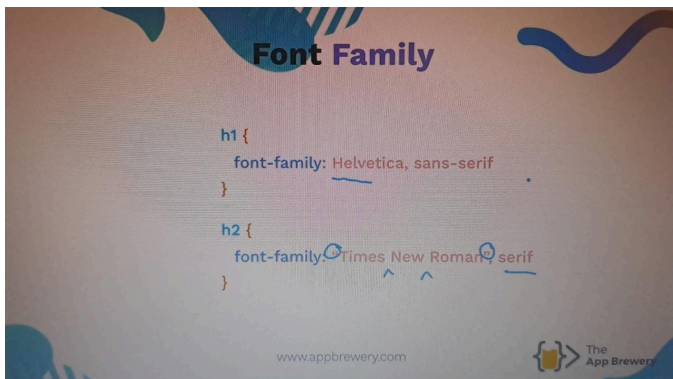
1x
2x



Font - weight:

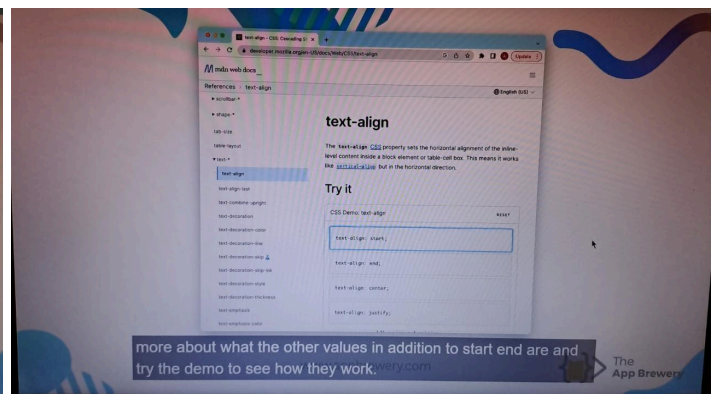
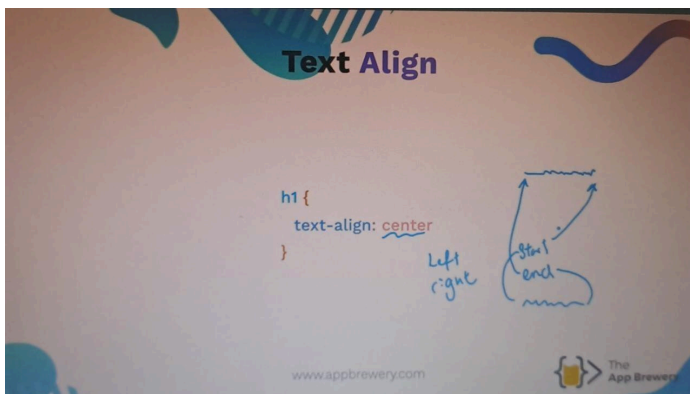


Font - family:



→ Google fonts.

<https://fonts.google.com/> for extraordinary fonts. From here copy link to <head>, font-family to css.
→ Check vid - 39 Project. (it's good)



<https://appbrewery.github.io/just-add-css/> - With and Without Css website.

Learning about Inspect:

Through Inspect we can look at Html and Stylesheet of websites. (Few Temporary changes can be done.)

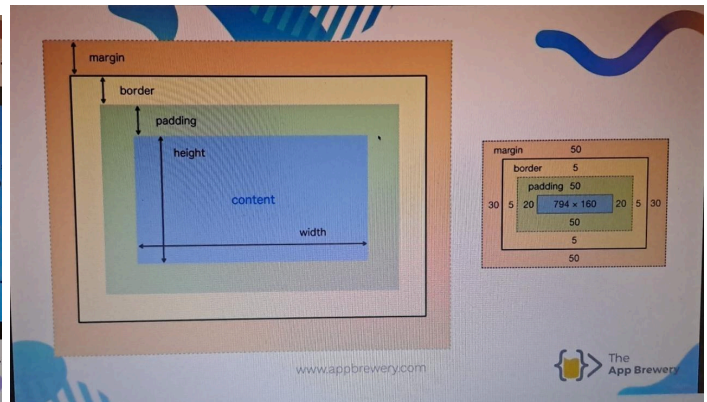
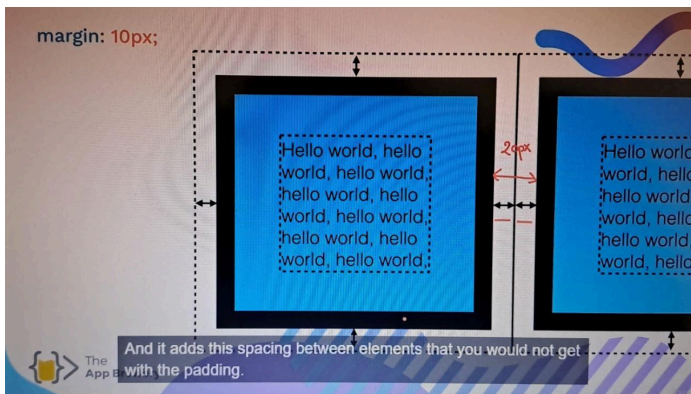
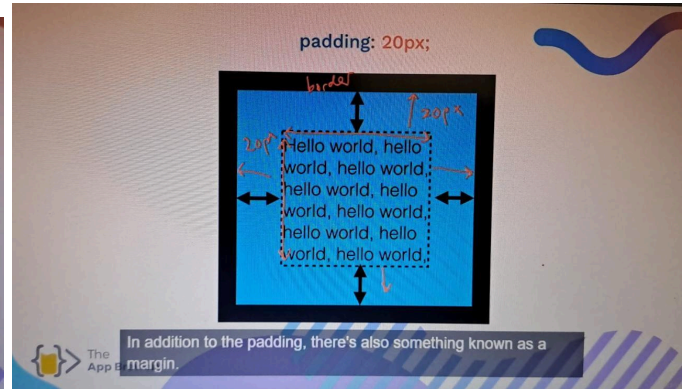
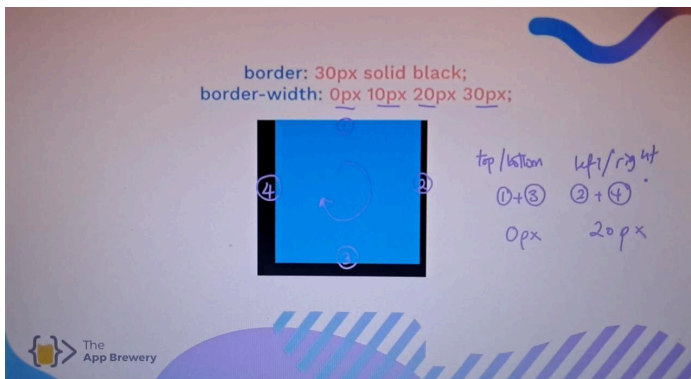
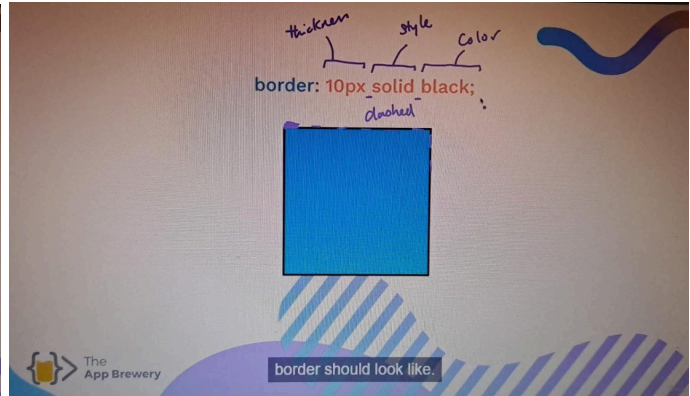
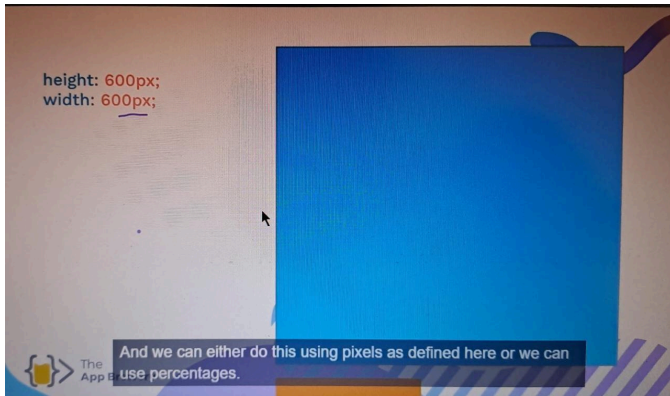
→ Go to Sources to look at actual Web files.

→ Go to the **computed** section in **elements**, to find out the **actual things applied** on the element.

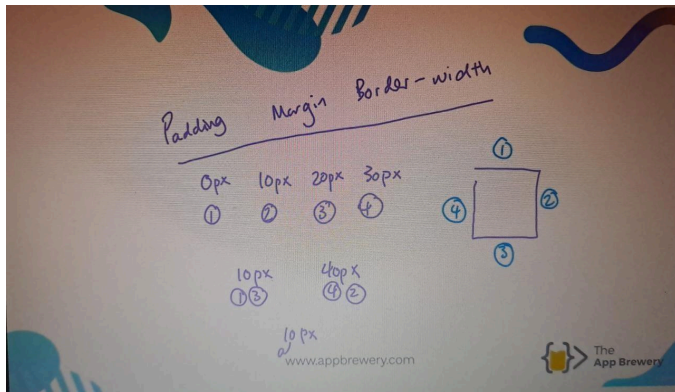
→ Go to 3 dots in Dev tools, select more tools, select **Css Overview**

<https://appbrewery.github.io/css-inspection/> - Inspect this website and answer it.

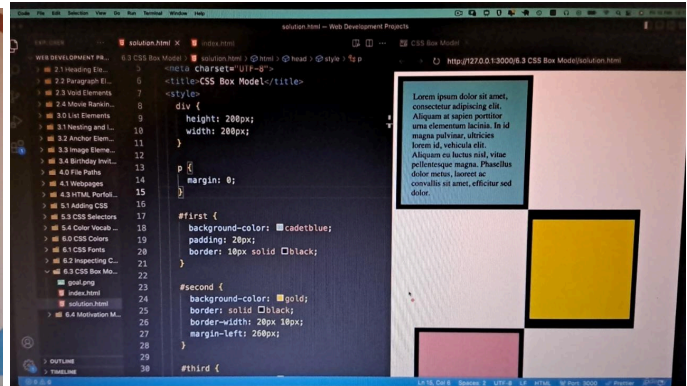
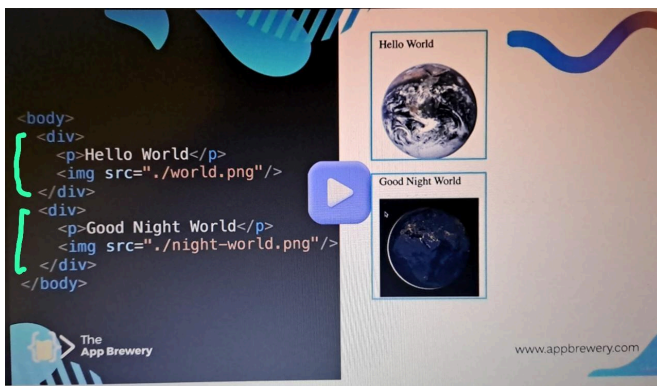
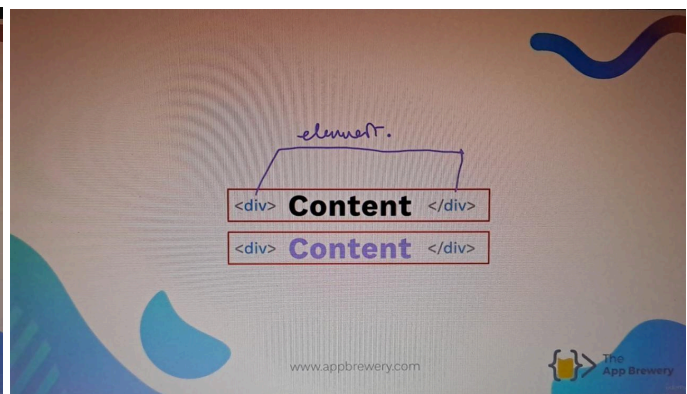
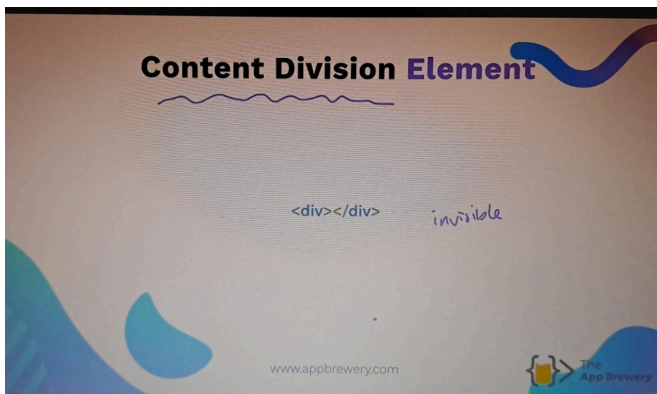
Box Model (Css):



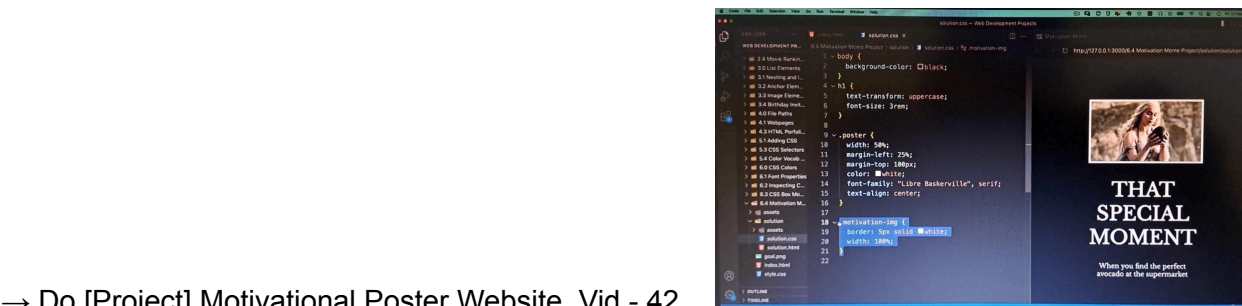
<https://appbrewery.github.io/box-model/> → Try changing the box model from things in here.



Content Division Element:



→ Try to complete the goal. Vid - 41.

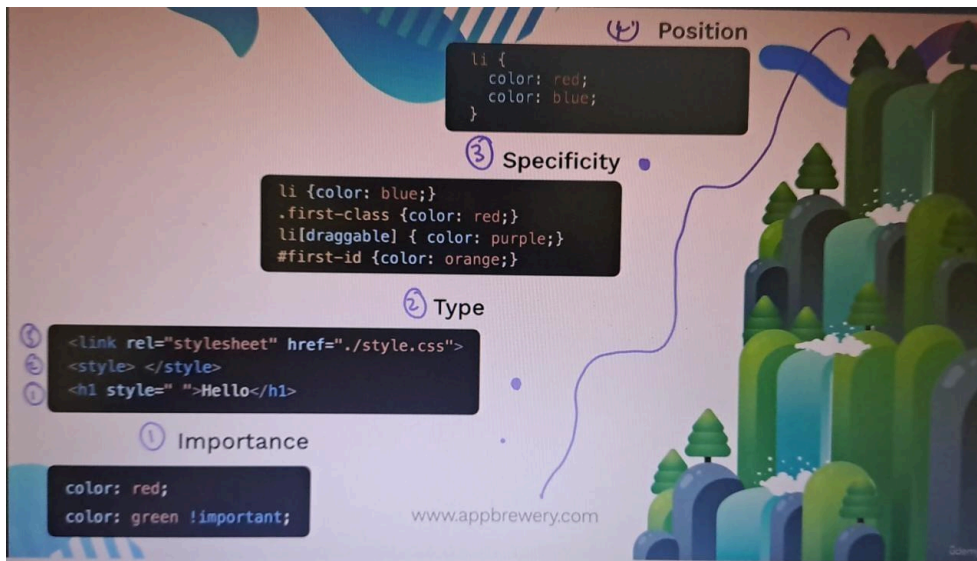


→ Do [Project] Motivational Poster Website. Vid - 42

It's good.

Section 7: Intermediate Css

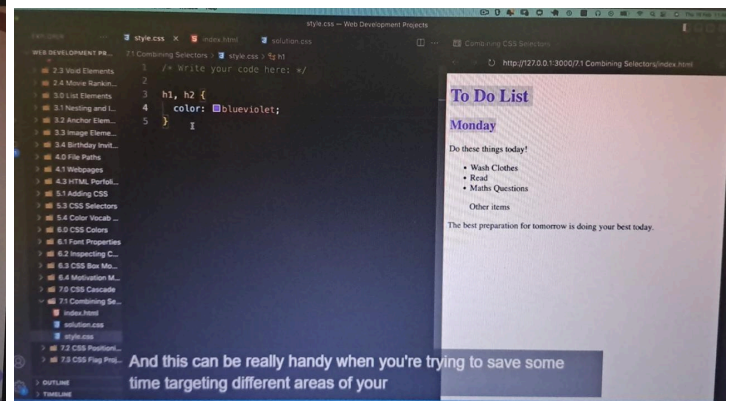
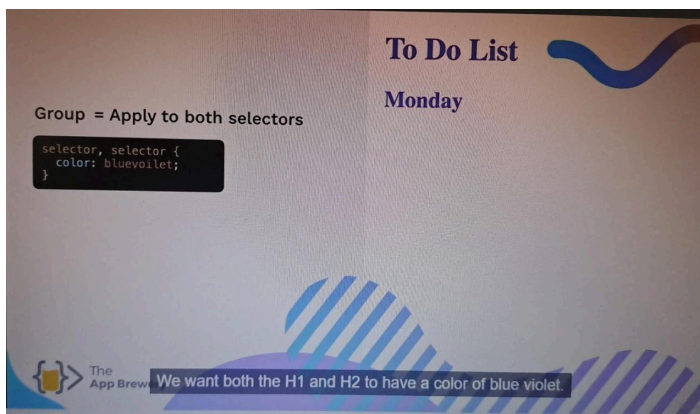
The Cascade - Specificity and Inheritance



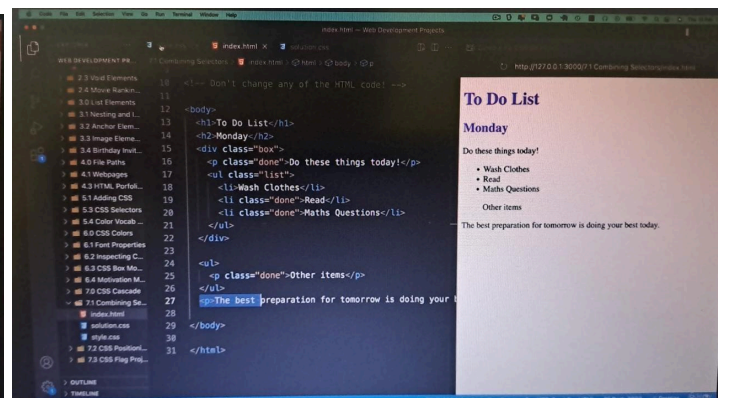
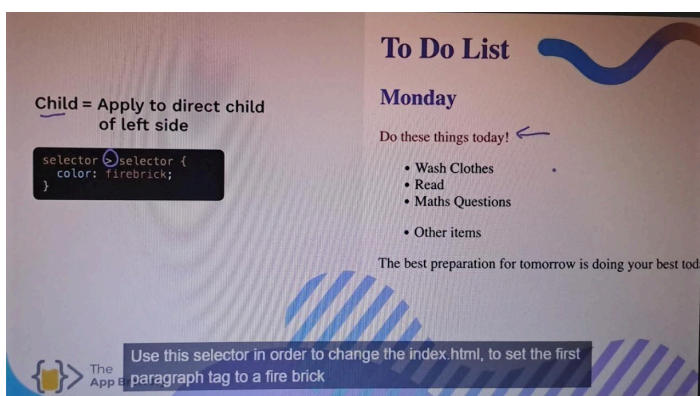
→ From up to down.

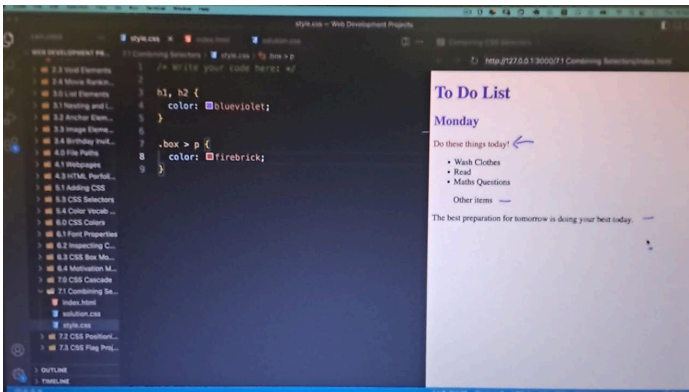
Combining Css selectors:

Group selector:

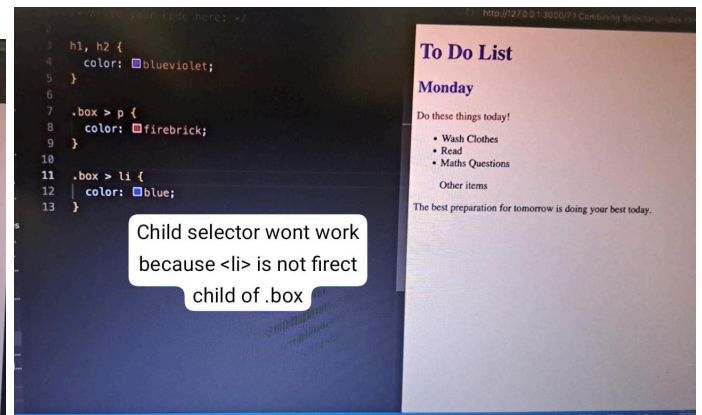
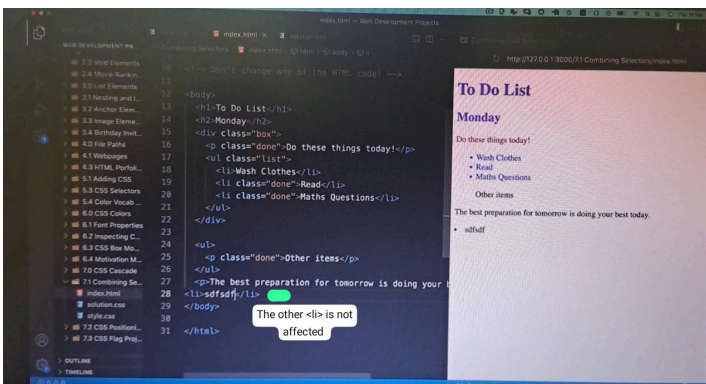
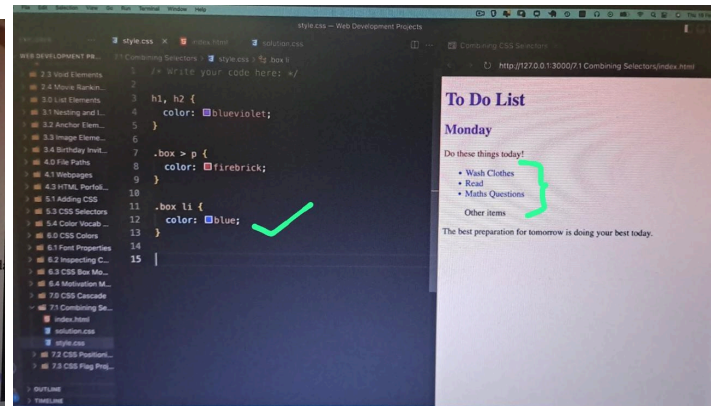
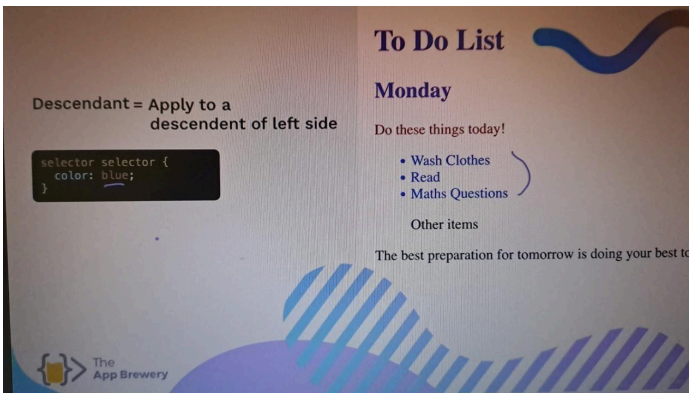


Child selector:

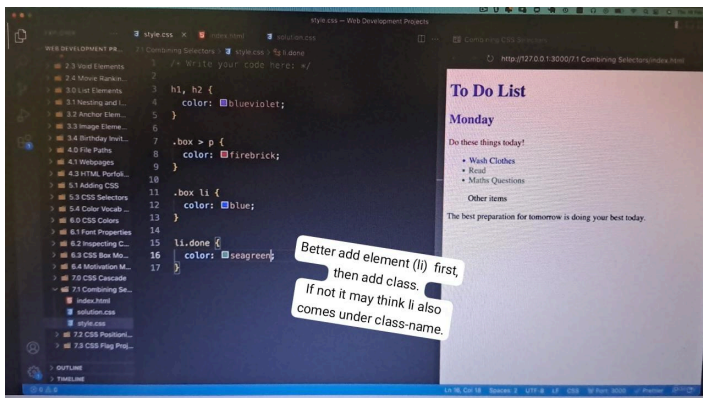
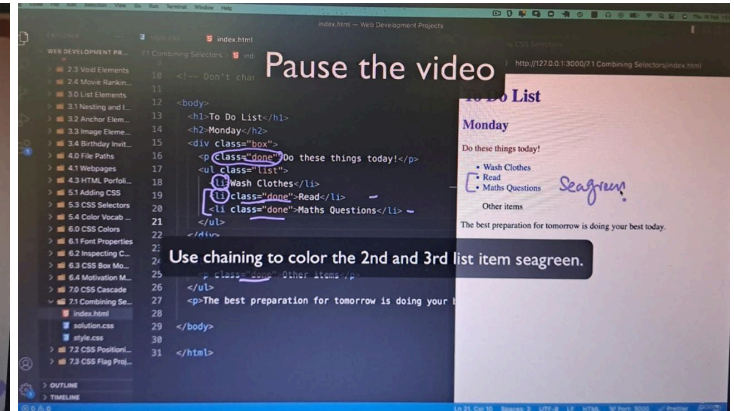
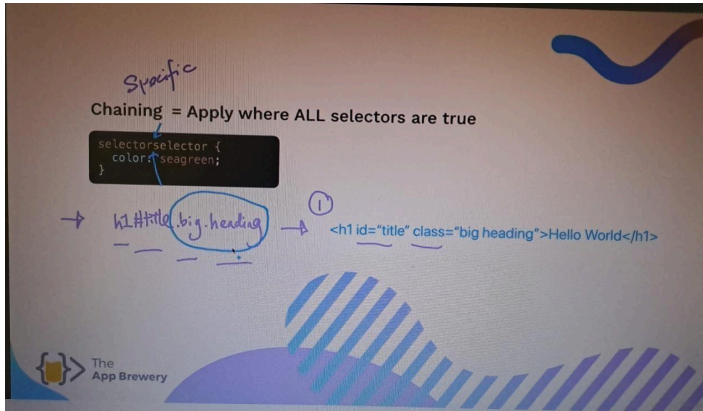




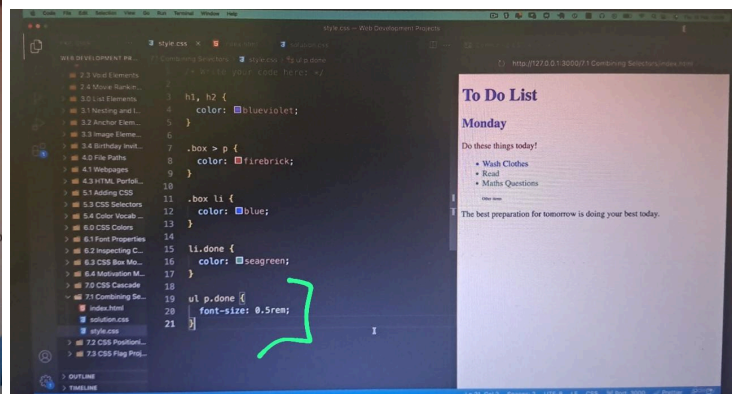
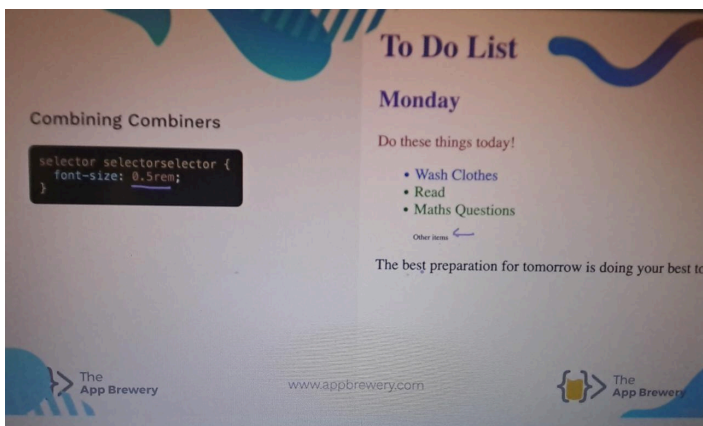
Descendant selector:



Chaining selector:

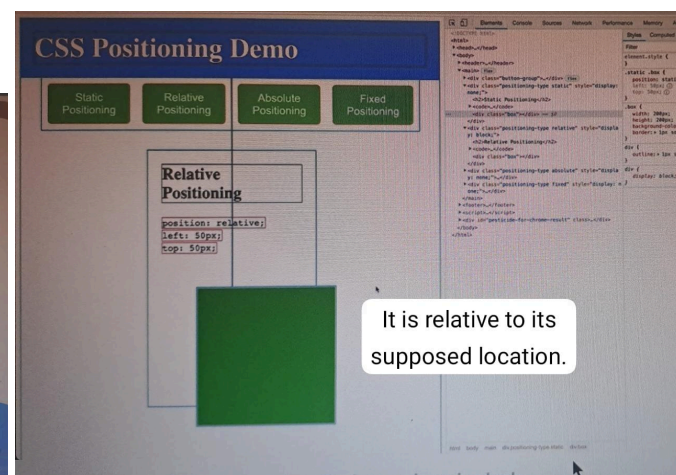
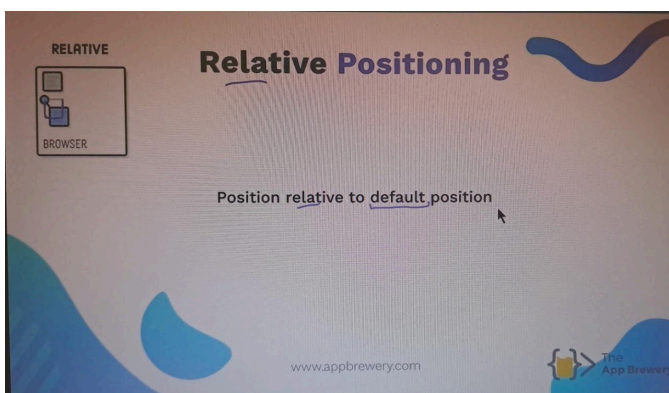
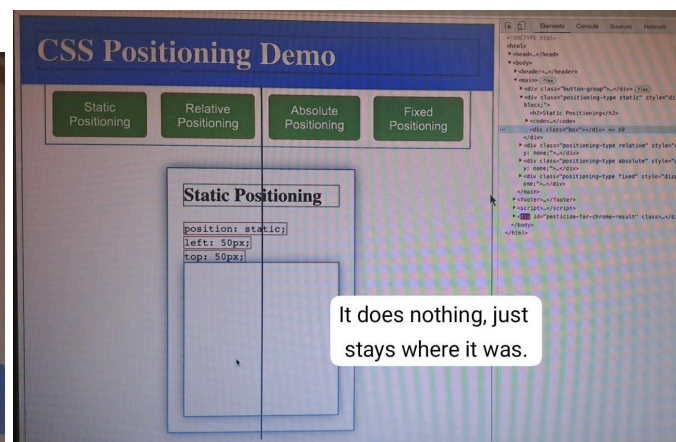
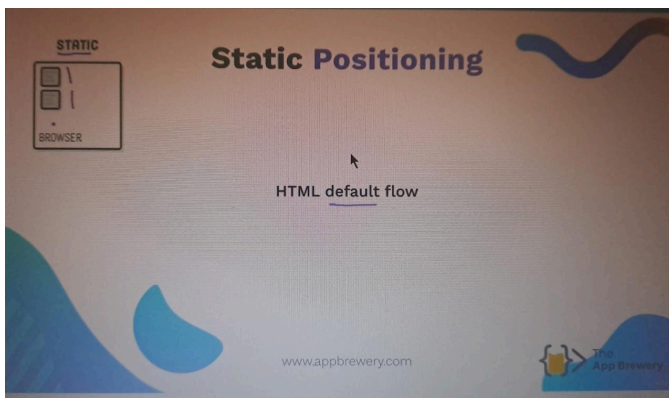
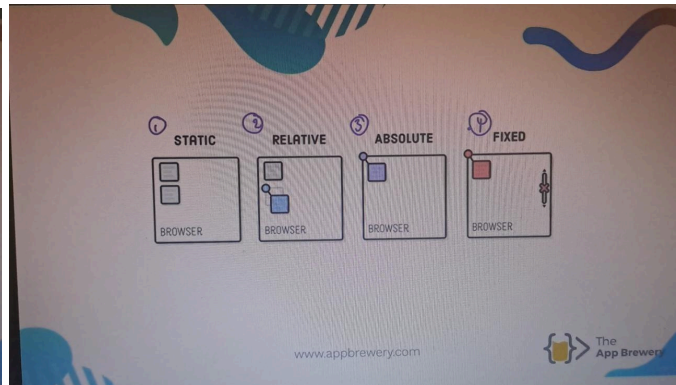
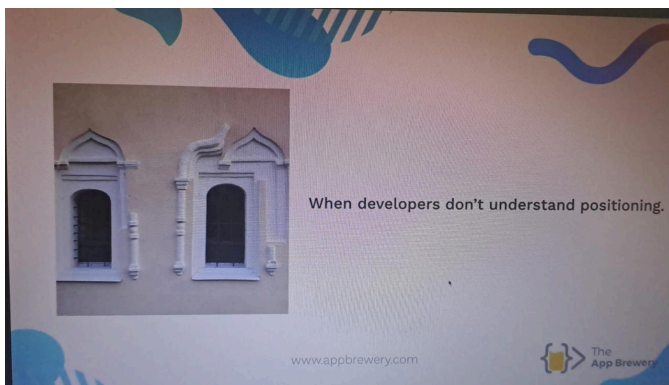


Finally Combining combinators:



Css Positioning:

<https://appbrewery.github.io/css-positioning/> - visit the site.



ABSOLUTE

Absolute Positioning

Position relative to nearest positioned ancestor
→ or top left corner of webpage

www.appbrewery.com

CSS Positioning Demo

Relative Positioning Absolute Positioning Fixed Positioning

Absolute Positioning

position: absolute; top: 50px; left: 50px;

CSS Positioning Demo

Static Positioning Relative Positioning Absolute Positioning Fixed Positioning

Absolute Positioning

```

<!DOCTYPE html>
<html>
<head>
<script>
</script>
</head>
<body>
<div class="button-group">
</div>
<div class="positioning-type static" style="display: none;">
</div>
<div class="positioning-type relative" style="display: none;">
</div>
<div class="positioning-type absolute" style="display: block; position: relative;">
<div class="positioning-type absolute">
</div>
</div>
<div class="positioning-type fixed" style="display: none;">
</div>
</body>
</html>

```

Z-Index

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CSS Positioning Demo

Static Positioning Relative Positioning Absolute Positioning Fixed Positioning

Absolute Positioning

```

position: absolute;
top: 50px;
left: 50px;

```

Negative z index to push it behind

FIXED

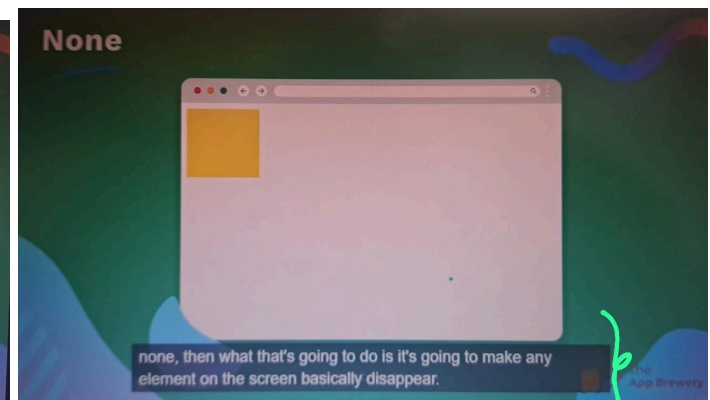
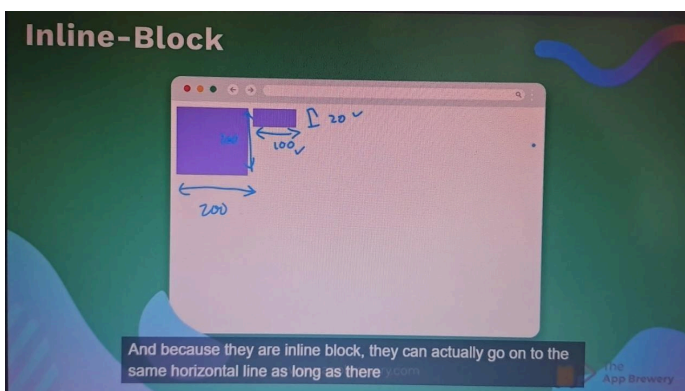
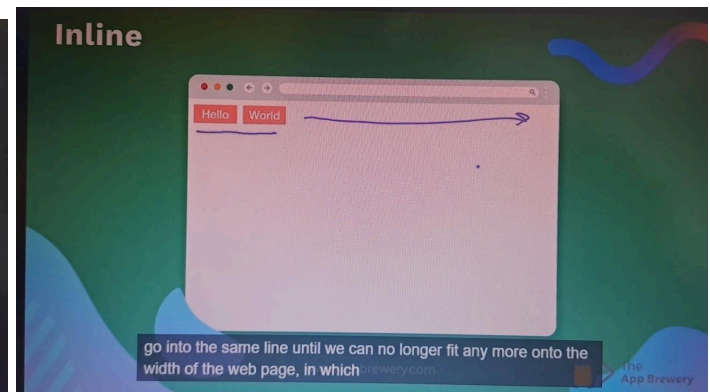
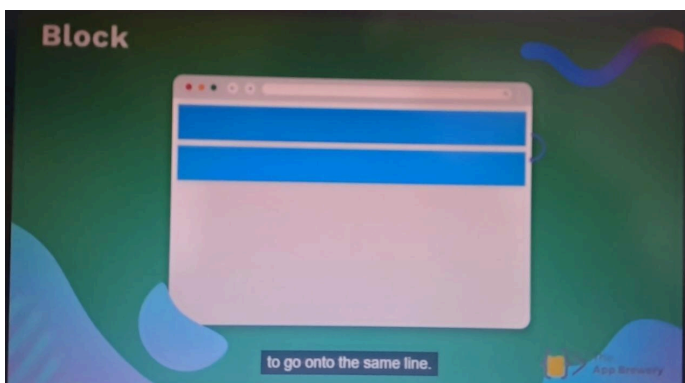
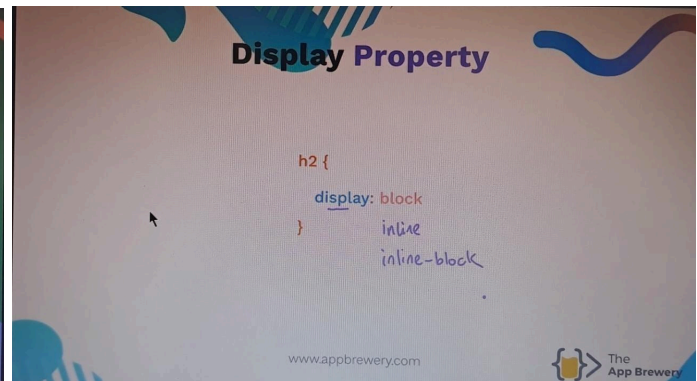
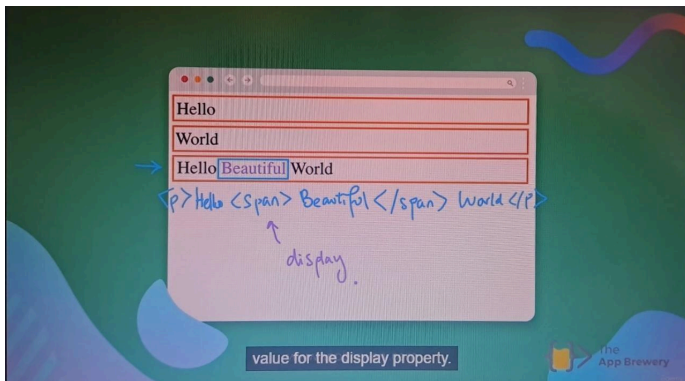
Fixed Positioning

Position relative to top left corner of browser window

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Section 8: Adv Css

Display property:



- <https://appbrewery.github.io/css-display/> - For more understanding.
- Block elements **can change** height and width, they **occupy** the whole right space.
- Inline **cannot change** height and width, they **don't occupy** the whole right space.
- Inline-Block can **change** H and W and also **not occupy** the whole right space.

Float property:

The App Brewery

CSS Float

Wrapping text using float and clear

And the main thing that this allows us to do is to wrap text around a particular element.

```
<img />
<p>text...</p>
```

```
img {
  float: left;
}
```

float left

Lorem ipsum dolor sit amet, in velit orci lectus eget diam, dolor. Et laoreet commodo molestie interdum accumsan, nonummy sed dis, dui ut feugiat interdum aenean ut sed, leo enim nam ipsum morbi, et sed.

We're not actually setting anything on the text here.

So take up the full width and then so on and so forth.

float left

float right

And now these two elements are now taken out of the normal flow and they're actually floating around

We need to achieve this

the footer to be free of this floating image, then we need to use a related property which is called **Clear**

```
<img />
<p>text...</p>
<footer>Copyright App Brewery</footer>
```

```
img {
  float: left;
}
footer {
  clear: left;
}
```

Copyright App Brewery

And this will end up looking something like this.

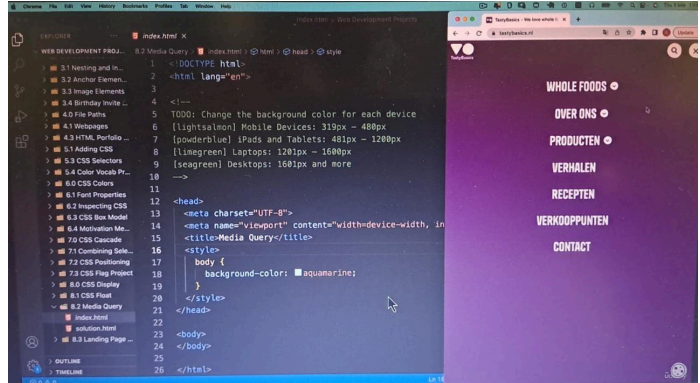
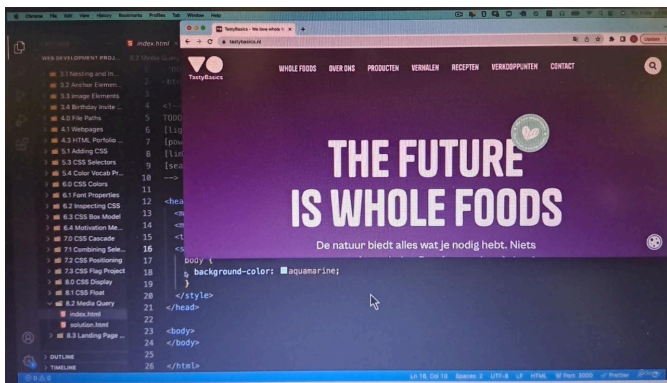
float left

float right

clear: both

We can achieve something like this.

How to make responsive websites?



- 1 Media Queries
- 2 CSS Grid To make it responsive, we can use.
- 3 CSS Flexbox
- 4 External Frameworks e.g. Bootstrap

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Media Query

```
@media (max-width: 600px) {  
  /* CSS for screens below or equal to 600px wide */  
}
```

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CSS Grid

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CSS Grid

```
.grid-container {  
  display: grid;  
  grid-template-columns: 1fr 1fr;  
  grid-template-rows: 100px 200px 200px;  
  gap: 30px;  
}  
  
.first {  
  grid-column: span(2);  
}  
  
.card {  
  background-color: blue;  
}
```

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CSS Flexbox

```
<div class="flex-container">  
  <div class="first card"></div>  
  <div class="second card"></div>  
  <div class="card"></div>  
  <div class="card"></div>  
</div>
```

```
.flex-container {  
  display: flex;  
}  
  
.card {  
  background: blue;  
  border: 30px solid white;  
  height: 100px;  
  flex: 1;  
}  
  
.first {  
  flex: 2;  
}  
  
.second {  
  flex: 0.5;  
}
```

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Bootstrap Framework

```
<div class="container">  
  <div class="row">  
    <div class="card col-6">  
      Card  
    </div>  
    <div class="card col-2">  
      Card  
    </div>  
    <div class="card col-4">  
      Card  
    </div>  
  </div>  
</div>
```

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Looking at Vid - 50 resources - really helps to understand these. 🚀

Media Query:

Media Query

breakpoint

```
@media (max-width: 600px) {  
  h1 {  
    font-size: 15px;  
  }  
}
```

width desktop website.

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index.html x

```
<div>  
  <div>  
    <div>  
      <div>  
        <div>  
          <div>  
            <div>  
              <div>  
                <div>  
                  <div>  
                    <div>  
                      <div>  
                        <div>  
                          <div>  
                        </div>  
                      </div>  
                    </div>  
                  </div>  
                </div>  
              </div>  
            </div>  
          </div>  
        </div>  
      </div>  
    </div>  
  </div>  
</div>
```

styles.css x

```
div {  
  background-color: blue;  
  height: 200px;  
  width: 200px;  
}
```

@media (max-width: 600px) {
 div {
 height: 100px;
 width: 100px;
 }
}

use of the screen real estate and it's scaled to fit to that screen size.

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Max Width

override

```
@media (max-width: 600px) {  
  h1 {  
    font-size: 15px;  
  }  
}
```

and we provide an override to the default styling.

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Min Width

max width 600 min width 2400px

```
@media (min-width: 600px) {  
  h1 {  
    font-size: 15px;  
  }  
}
```

So you're essentially targeting different screen sizes and providing different styles for those different

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Combine

600px 900px

```
@media (min-width: 600px) and (max-width: 900px) {  
  /* Styles for screens between 600px and 900px */  
}
```

system by adding that and keyword between these two breakpoints.

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Device

Print

```
@media screen (orientation: landscape) {  
  /* Styles for landscape orientation */  
}
```

But we will find it out there in wild

Not recommended to use

It's just saying targeting screens or targeting print.

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Using media queries

Media queries allow you to apply CSS styles depending on a device's general type (such as print vs. screen) or other characteristics such as screen resolution or browser viewport width. Media queries are used for the following:

- To conditionally apply styles with the CSS `media` and `supports` at-rules
- To target specific media for the `display`, `width`, `height`, and other `viewport` elements with the `viewport` attribute
- To test and monitor media states using the `media` and `supports` at-rules

Note: The examples on this page use CSS's `media` for illustrative purposes, but the basic syntax remains the same for all types of media queries.

Syntax

A media query is composed of an optional media type and any number of media feature expressions, which are separated by `and` or `not` keywords. Media queries are used for the following:

For more on media queries and all of the different properties that you can change using media queries.

Media Query

```
<!DOCTYPE html>  
<html lang="en">  
  <head>  
    <meta charset="UTF-8">  
    <meta name="viewport" content="width=device-width, initial-scale=1.0">  
    <title>Media Query</title>  
    <style>  
      body {  
        background-color: aquamarine;  
      }  
    </style>  
  </head>  
</html>
```

Try to do Vid 51 Resource. Its good

this kind of powder blue and finally mobile devices which has this salmon, orange or red.

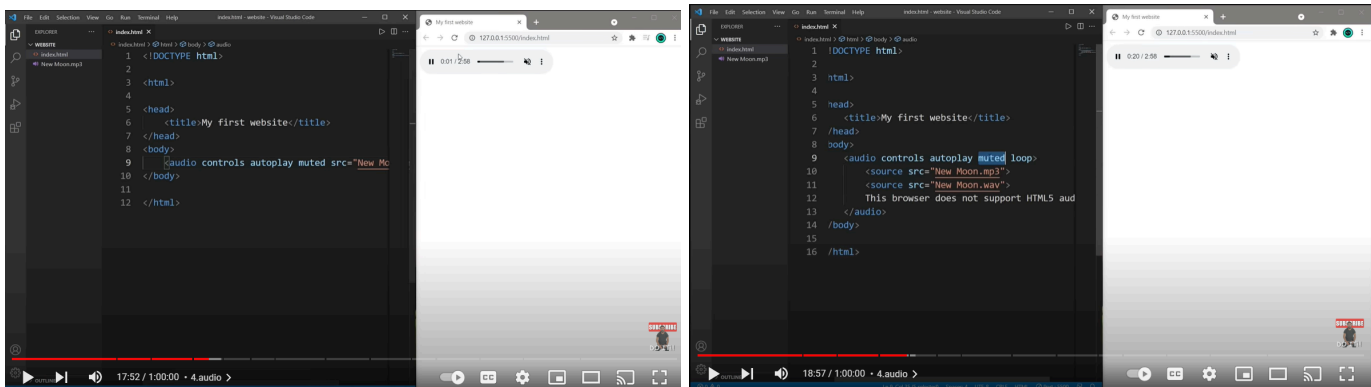
Look at Vid 51 Resource. Improves understanding for Media Query.
Web design agency Website [Project] - Very nice.

→ This was up to Advanced CSS.

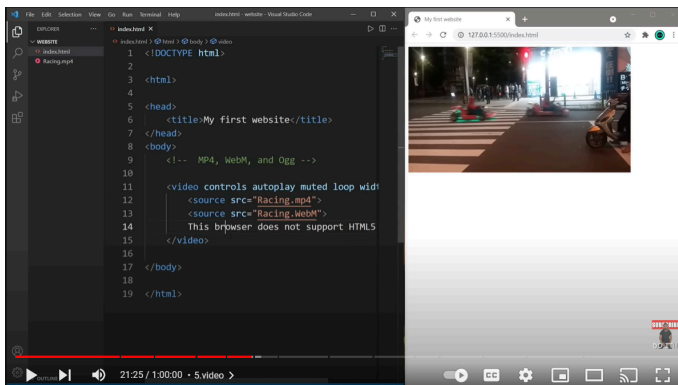
Additional points :

Bro code :

<audio controls autoplay muted loop src= "New moon.mp3">

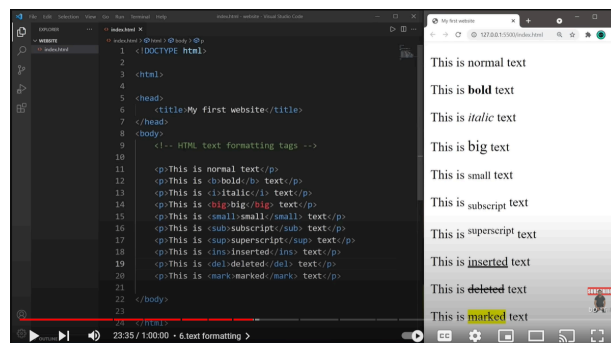


<video controls autoplay muted loop src ="Racing.mp4" width="500">



<source src="Racing.mp4"><source src="Racing.WebM">

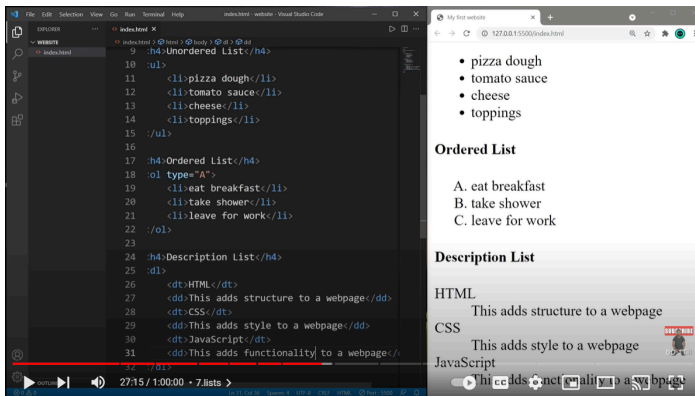
These are for backup, in case the browser doesn't support html 5.



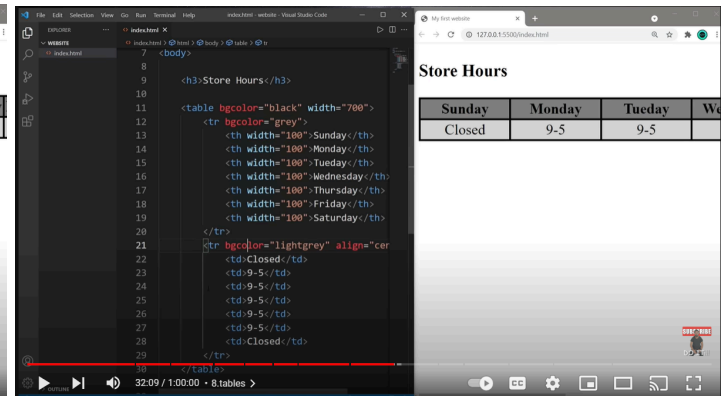
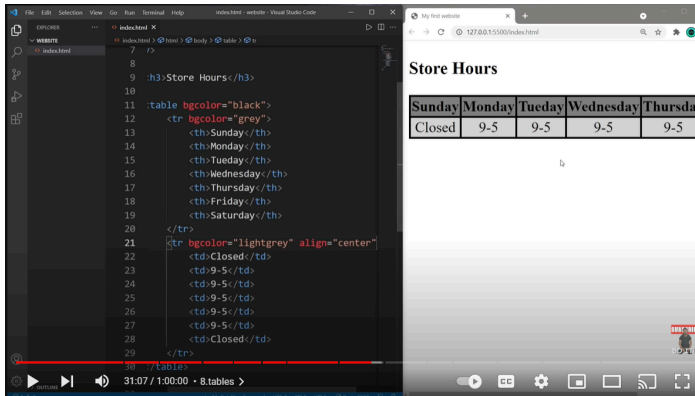
→ Text editing tags _

→ It's cool.

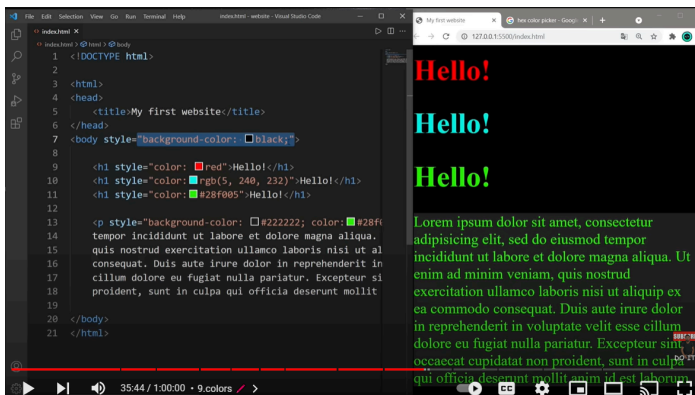
→ A new list - 'Description list.'



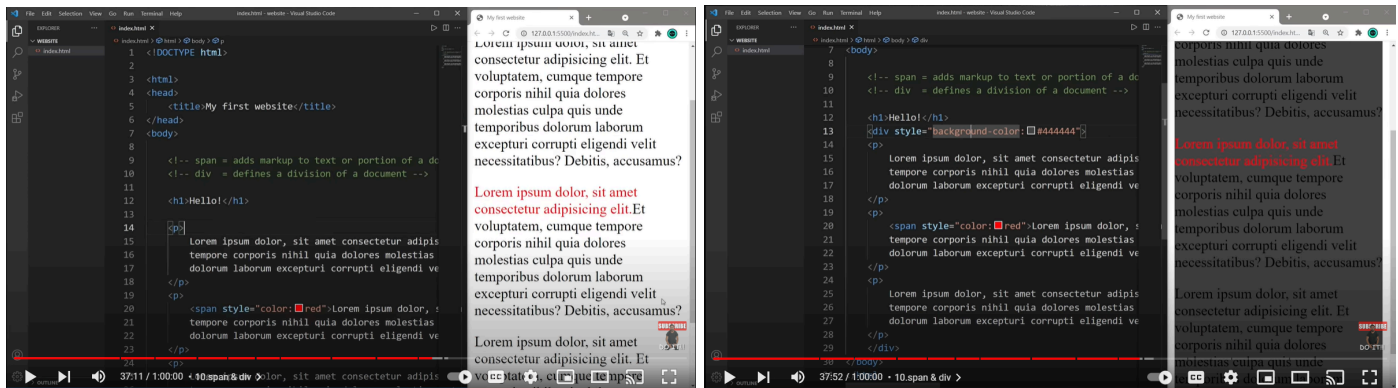
→ Tables



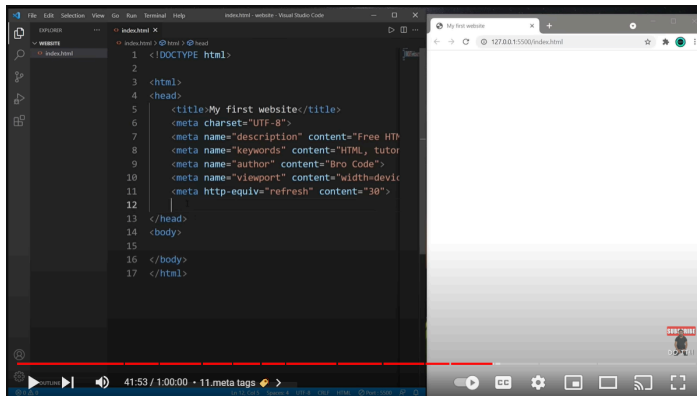
_ More randoms



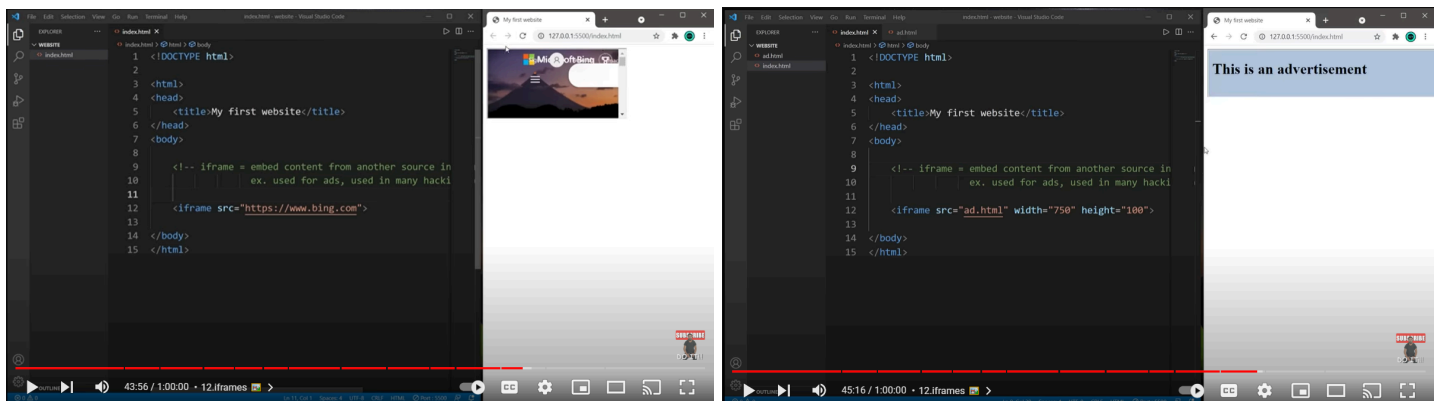
→ Span and div tags_



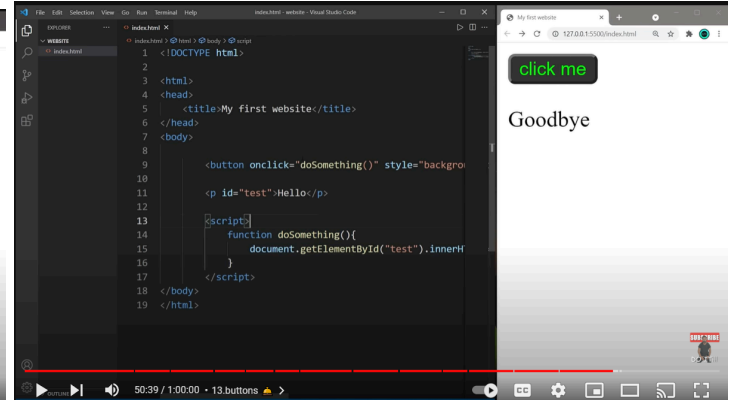
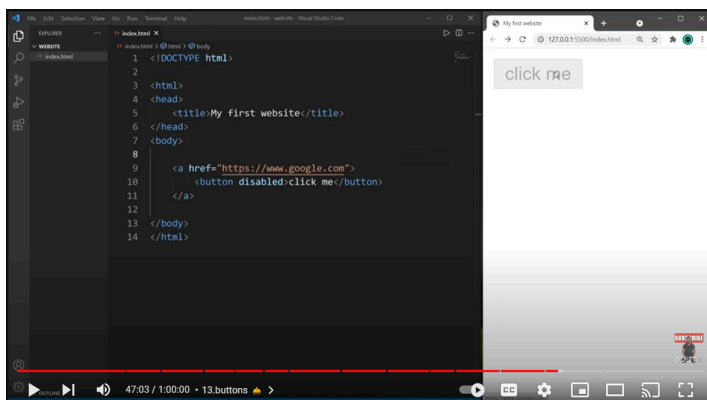
→ Meta tags_



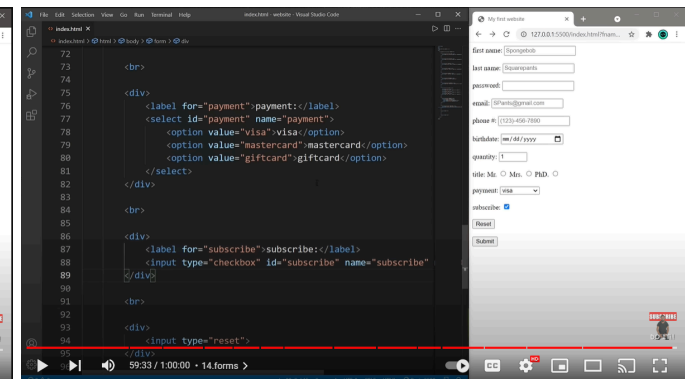
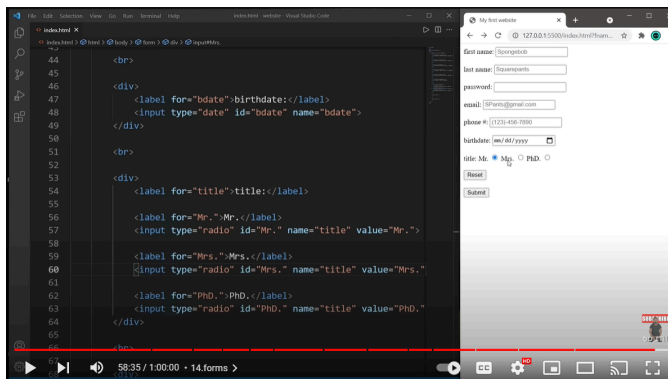
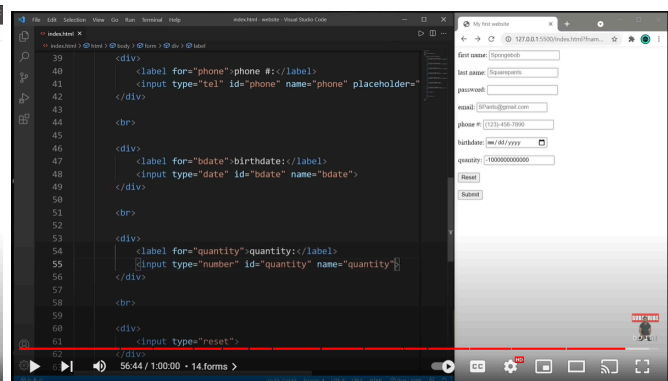
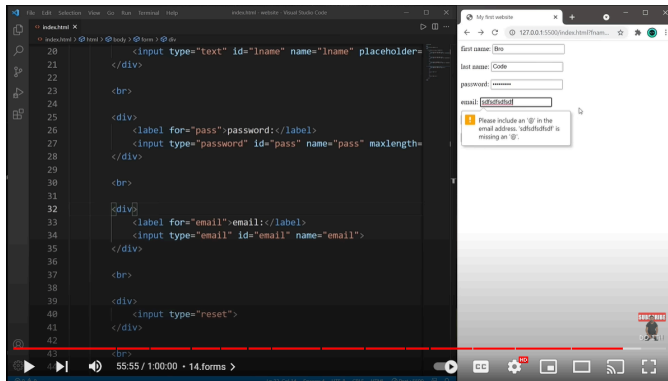
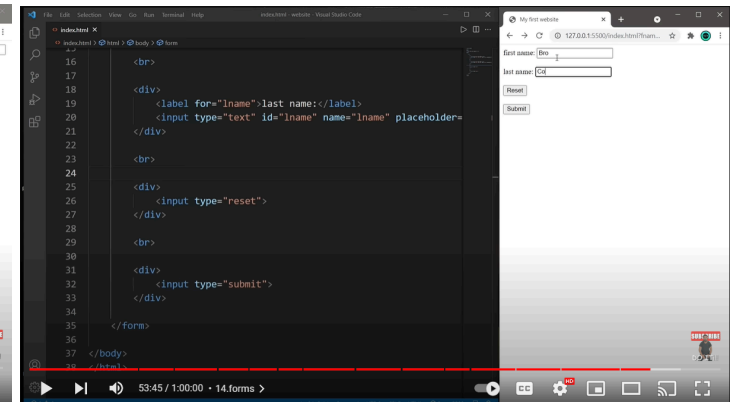
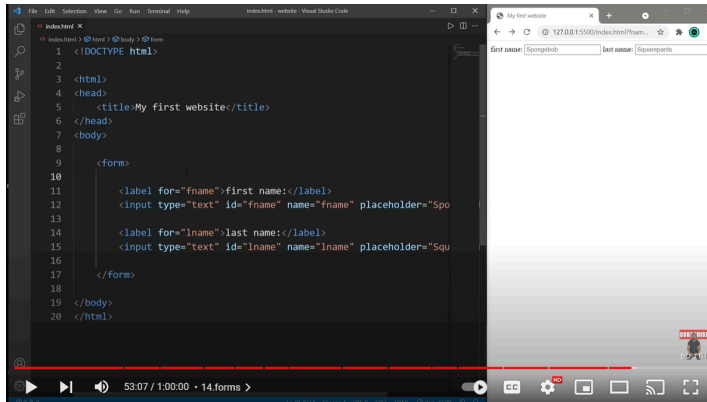
→ iframe_



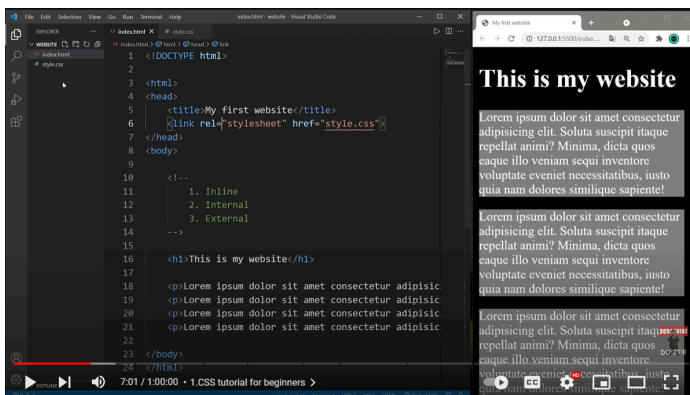
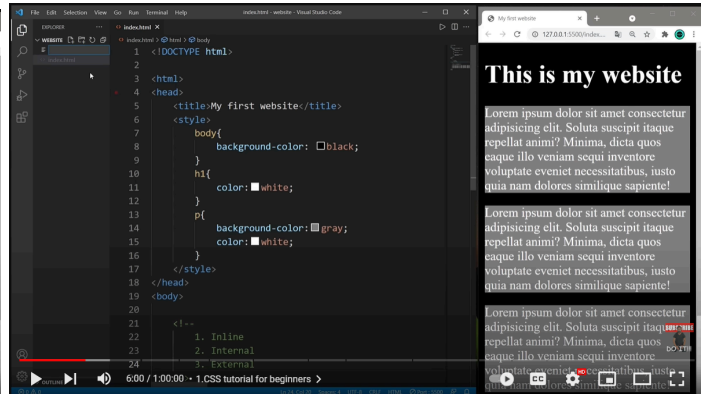
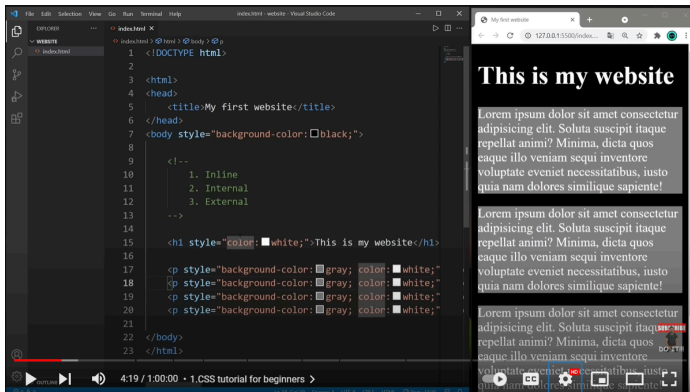
→ Button



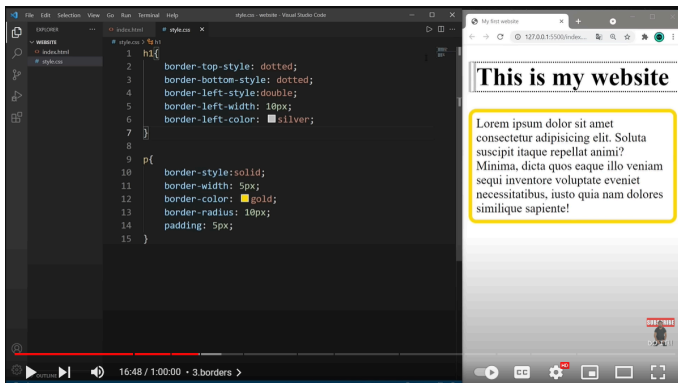
Tables



CSS_

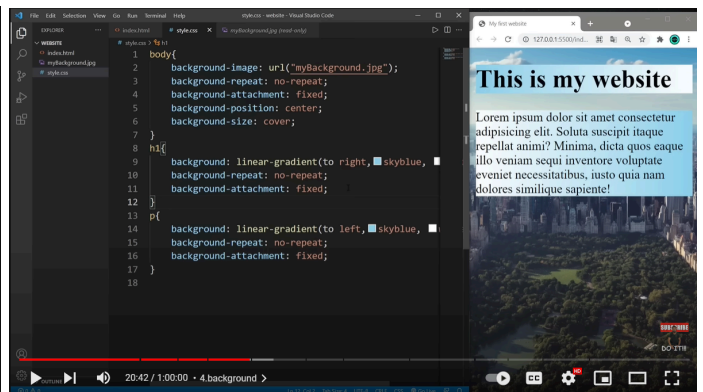
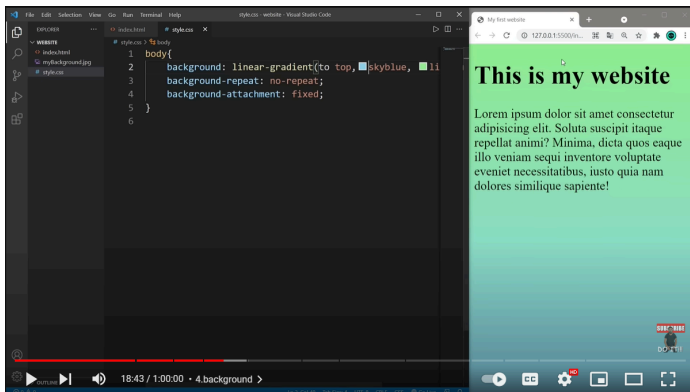


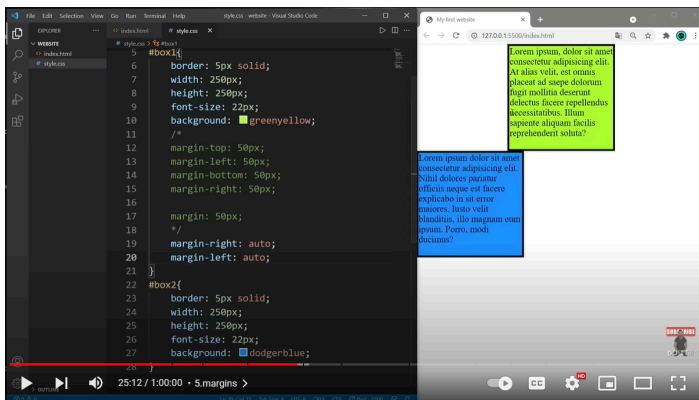
This was inline, internal and external css.



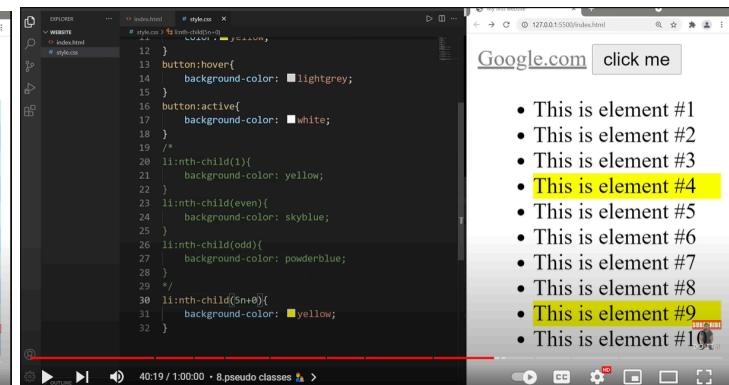
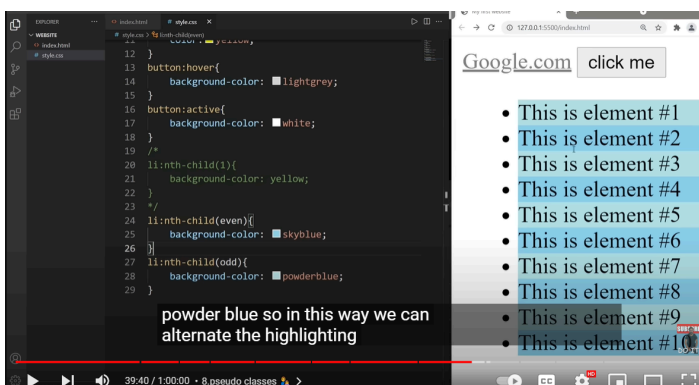
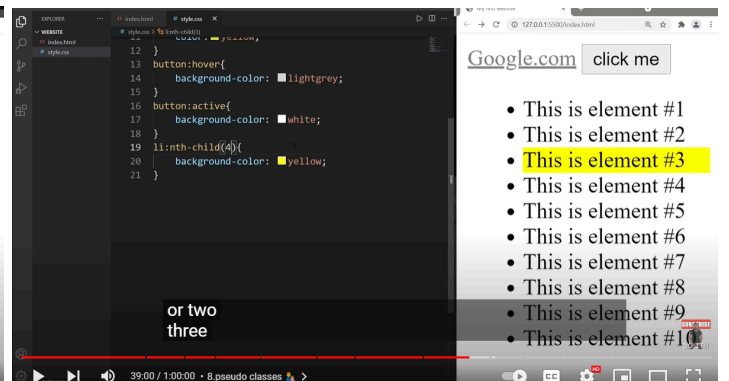
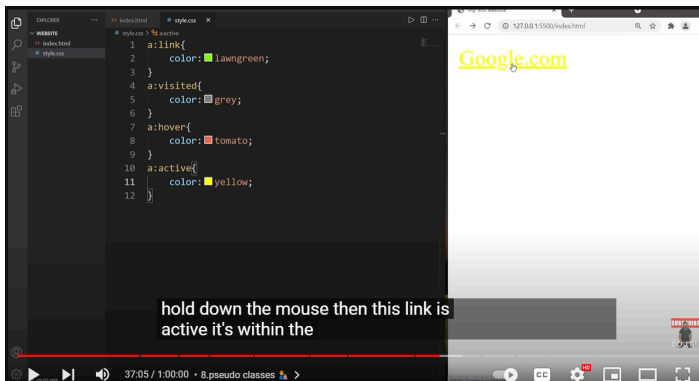
This is border styles.

Background:

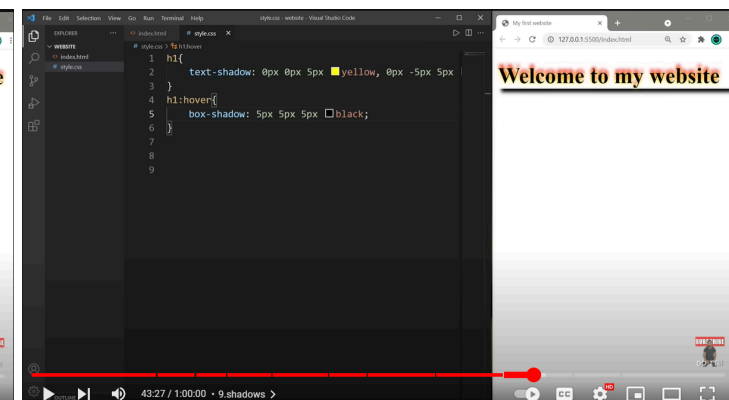
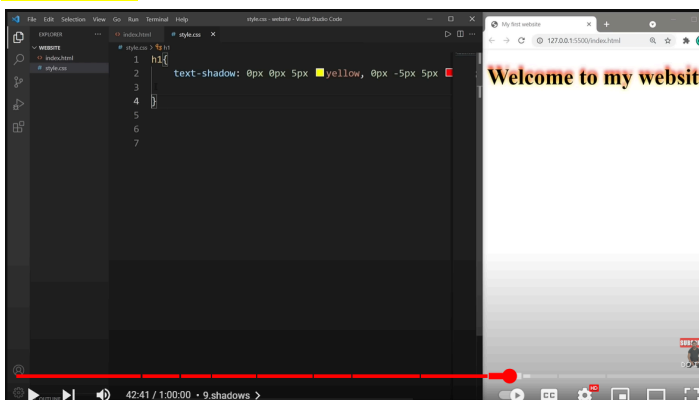




→ Next, pseudo classes

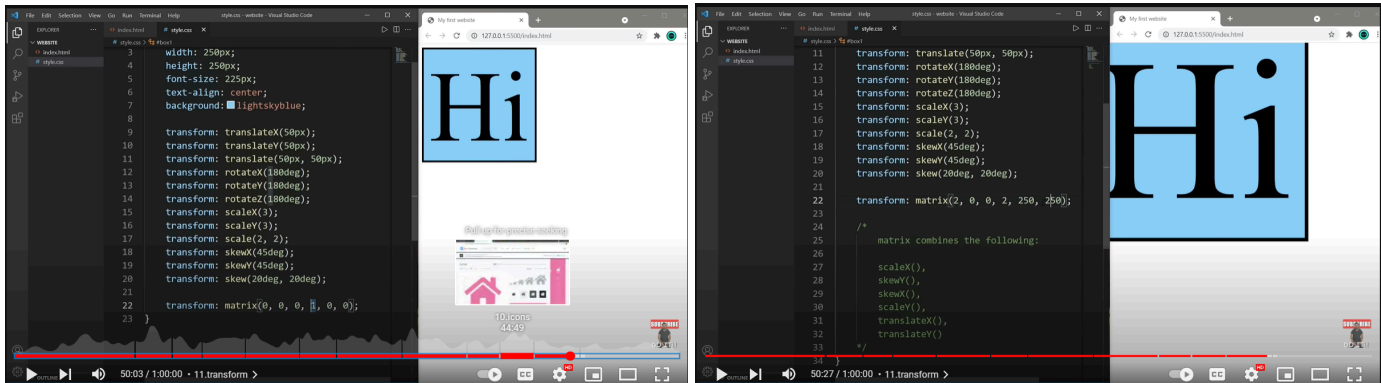


Shadows :

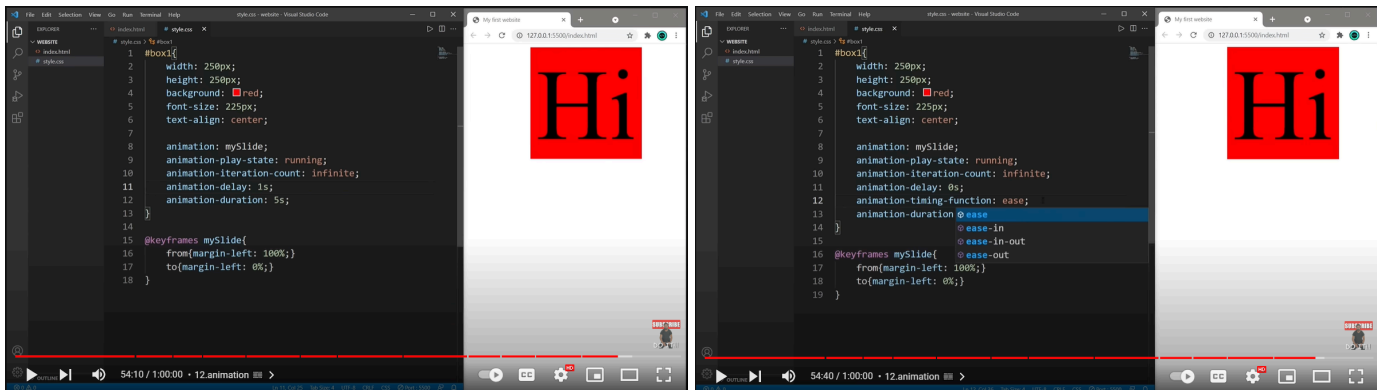


Font Awesome website - For "icons". → first import script then <i> tags for different icons.

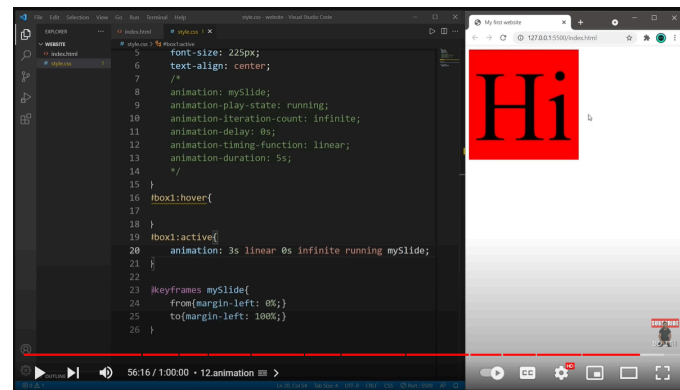
→ Transform.



Animations:

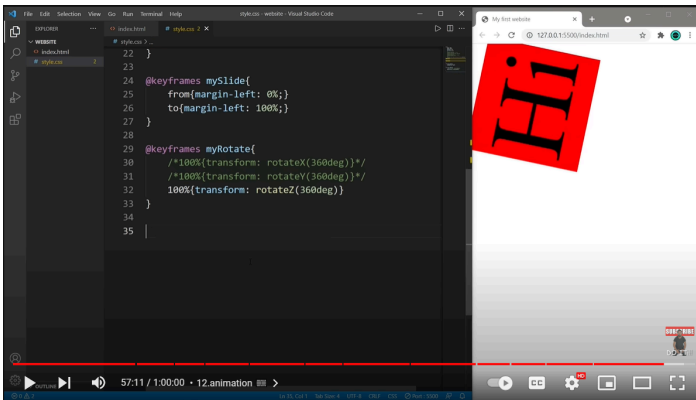


Ease : comes in speed then slowly slows down. Linear : comes in uniform animation.

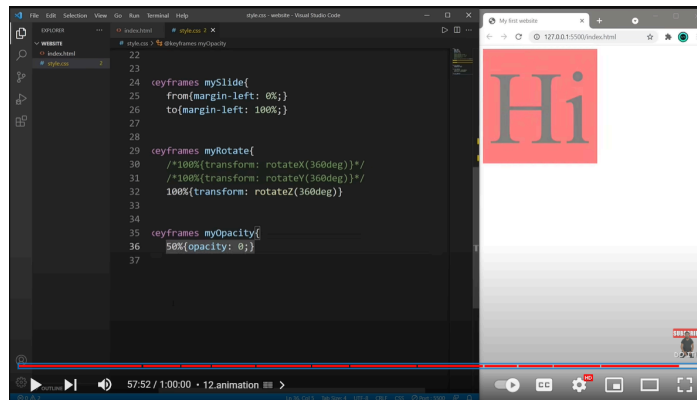


That line in active (means start on click) is a shortcut for all the about.

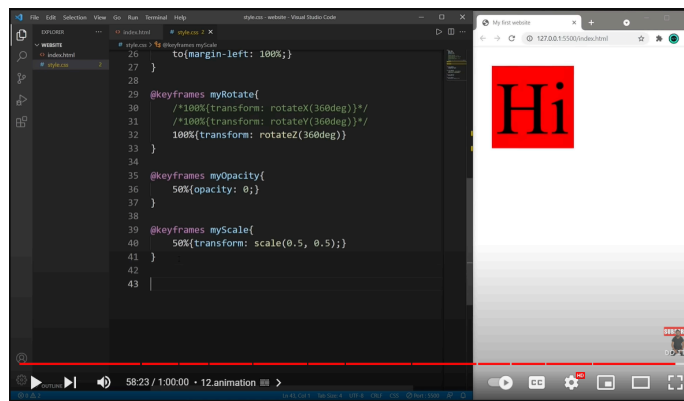
Now, let's try to make it rotate.



Then, the gradual fade in and fade out :



Now, the image will shrink and grow :



Now, color changing animation :

